Team 5

C373 FA

Problem statement

" How can blockchain-based decentralized applications (dApps) be leveraged to build a secure, transparent, and efficient e-commerce platform in Singapore that improves trust between buyers and sellers while reducing fraud and operational inefficiencies. "

Business Analysis – Key challenges in eCommerce

Payment scams and disputes

Payment scams and disputes causes financial losses and reducing trust between buyers and sellers.

Poor shipment visibility

Poor shipment transparency in e-commerce is caused by inaccurate information, lack of updates, and weak communication.

Fake reviews and unreliable ratings

Fake reviews occur when sellers manipulate ratings by paying or refunding users to post positive reviews, even without real product experience.

Fragmented systems and data silos

Disconnected systems create data silos that hinder customer insights, personalization, and business efficiency.

Counterfeit and unauthentic products

Lack of reliable product verification allows counterfeit goods to reach customers

Business Analysis – Blockchain Solutions

Payment scams and disputes

Smart contract escrow releases payment only after delivery

Poor shipment visibility

On-chain shipping updates ensure transparent tracking

Fake reviews and unreliable ratings

Only verified buyers can leave tamper-proof reviews

Fragmented systems and data silos

Ensures all parties access the same trusted and consistent data using shared ledger

Counterfeit and unauthentic products

Hashed product records prevent data manipulation

It improves trust between buyers and sellers by reducing fraud, disputes, and operational inefficiencies.
Overall, it enables a secure, transparent, and efficient decentralized e-commerce platform.

Blockchain platform

Platform	Key Characteristics	Decision	Justification
Hyperledger Fabric	• Private, permissioned network • Enterprise-focused • Practical Byzantine Fault Tolerance • No public gas fees • Not EVM compatible	Not Selected	Unsuitable for a public marketplace requiring open buyer/seller participation
Ethereum	• Public ledger, largest ecosystem • Solidity smart contracts • MetaMask native support • Strong decentralization & security • Higher gas fees (~$0.05+) • Lower throughput (~30 TPS)	Reference Platform	Used as security anchor via Polygon checkpoints; too expensive for frequent micro-transactions
Polygon	• Ethereum-compatible L2 solution • Proof-of-Stake consensus • ~7,000 TPS, ~$0.01 fees • EVM support (Solidity smart contracts) • MetaMask compatible • Checkpoint security via Ethereum mainnet	Selected	Primary platform – best balance of low fees, high throughput, and security for consumer e-commerce

Polygon was selected due to its faster transactions, lower fees, and full Ethereum compatibility, making it ideal for a public e-commerce marketplace, while Hyperledger Fabric is too restricted and Ethereum is too costly for frequent transactions.

Consensus mechanism comparison

Mechanism	Proof of Work (Pow)	Proof of Stake (PoS)	Delegated PoS / PBFT
How it works	Miners solve puzzles to validate blocks	Validators stake tokens to propose/verify blocks	Token holders vote for trusted validators (DPoS) or reach agreement via voting rounds (PBFT)
Pros	Very secure, proven model (used by Bitcoin).	Fast confirmation, low fees, scalable, and energy-efficient	Very fast and efficient, low energy use
Cons	Slow transaction times, high energy consumption, and expensive fees.	Security depends on fair distribution of stake, but Polygon anchors checkpoints to Ethereum for added trust.	Less decentralized — relies on a few trusted validators, which can reduce transparency.
Does it fit our problem statement ?	Not suitable, delays and high costs would worsen settlement inefficiencies and discourage adoption in fast-paced e-commerce	PERFECT, it reduces transaction fees, speeds up settlements, and ensures secure peer-to-peer transactions. Perfect for Singapore's high-volume marketplace.	Not suitable, better suited for private or enterprise blockchains, but not for open consumer marketplaces where decentralization builds trust.

Consensus recommendation: Proof of Stake (PoS)

Current Challenges	Why PoS ?
High transaction fees	PoS minimizes gas costs, Polygon PoS averages ~$0.01 per transaction, enabling affordable buyer-seller interactions.
Delayed settlements	Fast block times (~2s) ensure near-instant order confirmation and payout.
Fraud and data manipulation	Ethereum-anchored checkpointing secures transaction history and product authenticity
Limited trust between users	Decentralized validator network removes reliance on intermediaries and builds peer-to-peer trust.
Scalability for high-volume e-commerce	Polygon PoS supports ~7,000 TPS, ideal for Shopee-style marketplaces with frequent transactions.

Conclusion:
PoS on Polygon delivers the speed, cost-efficiency, and security needed to power a decentralized e-commerce dApp that solves real-world problems in Singapore's digital economy.

Architecture Design Comparison

Architecture Design	Layer 2 Solutions	Sidechains	Hybrid Models
Description	Built on top of the main blockchain (Polygon) to provide scalability, reduced transaction costs, and faster finality.	Independent blockchains running alongside the main chain, connected via a two-way bridge.	Combining Layer 1 (main chain) and Layer 2 (sidechains or Layer 2 solutions) for scalability and security.
Advantages	- Enhanced scalability- Reduced fees - Fast transaction finality	- Customizability - High throughput- Lower transaction fees	- Combines scalability with security - Flexibility in design
Challenges	- Potential decentralization trade-offs - Complexity in development	- Security concerns if not properly validated - Requires bridges for cross-chain communication	- Complexity in integration - Higher development overhead
Best Use Case	E-commerce platforms needing high throughput and low fees for frequent transactions.	Platforms needing specialized features or private transactions while still leveraging the main chain's security.	Complex platforms with multiple transaction types requiring both security and scalability.

Architecture Design Chosen: Layer 2 Solution with Polygon zk-Rollups

What is it?

- **Polygon zk-rollups** are a specific type of **Layer 2 scaling solution** that work by processing transactions off the main Ethereum blockchain (Layer 1) but still ensure that the data is securely linked to it.
- **ZK** (Zero-knowledge), refers to cryptographic proofs used to confirm the correctness of transactions without revealing sensitive data.

How does it work?

- Rollups bundle multiple transactions together and process them **off-chain**.
- Transactions are then bundled into a single batch, and only a **cryptographic proof** (called a **ZK-proof**) is submitted to Ethereum.

Why this?

- It brings the benefits of **scalability** and **security** to blockchain applications, especially in scenarios where frequent, small transactions occur, such as e-commerce.
- They help solve the problem of high fees and slow transaction speeds, making blockchain technology more accessible and practical for businesses.

Token Model Recommendation: ERC-1155

Key Challenges in Singapore’s E-Commerce Sector	How ERC-1155 solves it
Counterfeit products and lack of authenticity	Authenticity & Provenance: Each product or batch can be represented by a unique token ID, ensuring verifiable listings and reducing counterfeit risks
High transaction fees and inefficiencies	ERC-1155 supports batch transfers, which lower gas fees, critical for high-volume marketplaces with frequent transactions.
Limited traceability in supply chains	Semi-fungible tokens allow tracking of product units across the supply chain, improving transparency.
Disputes over payments and delivery	Smart contracts built on ERC-1155 can enforce escrow, refunds, and settlements automatically, reducing reliance on intermediaries.

Conclusion:
ERC-1155 enables secure, transparent, and efficient product representation in our Shopee-style dApp, directly addressing fraud, authenticity, and cost challenges in Singapore’s e-commerce ecosystem.

Security recommendation: Smart contract security & audits

Smart Contract Security & Audits Address These Issues

- **Tamper-proof logic**: Smart contracts automate escrow, refunds, and payouts without intermediaries.

- **Fraud prevention**: Audits detect vulnerabilities before deployment, reducing risk of exploits.

- **Trust & transparency**: Verified contracts ensure product listings and transactions cannot be manipulated.

- **Consumer confidence**: Independent audits (e.g., CertiK, OpenZeppelin) build credibility with buyers and merchants.

- **Regulatory alignment**: Secure contracts support compliance with Singapore's digital payment token guidelines.

Conclusion:
Smart contract security and audits are essential to deliver a trustworthy, fraud-resistant, and transparent e-commerce dApp on Polygon PoS, directly solving the pain points of Singapore's fast-growing digital marketplace.

Smart contract Features Listing with Short Description

Basic

Feature	Description
Product Registration	Sellers register products with product details
Product Information	Stores product details, and metadata on-chain
Purchase Function	Buyers purchase products using ETH
Stock Management	Automatically reduces stock after purchase
Product Update	Creator can update product details
Event Logging	Emits events for product registration and purchase

Advanced

Feature	Description
Payment Automation (Escrow)	Funds locked in smart contract until delivery is confirmed
Shipping & Delivery Tracking	On-chain tracking updates linked to delivery milestones
Auto Payment Release	Payment released only after delivery confirmation
Loyalty Rewards Token	ERC-20 tokens issued after successful purchases
Token Redemption	Tokens used for discounts or platform incentives

Demo time!

Thank you